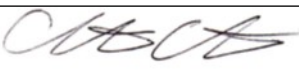


Guideline Title: Pediatric Ventricular Tachycardia with Pulse

Guideline Number: GUID-3203-PED018

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** Ensure prompt identification and provide intervention to restore a stable cardiac rhythm with adequate cardiac output and.

PATHOPHYSIOLOGY: Sustained ventricular tachycardia (VT) carries the risk of deteriorating into Ventricular Fibrillation (VF). Electrolyte imbalances, toxins, or cardiac ischemia can cause VT. It is uncommon in the pediatric patient. Rate may vary from near normal to > 200 bpm. Most children who develop VT have underlying heart disease (or have had surgery for heart disease), long QT syndrome, or myocarditis/ cardiomyopathy.

Conditions and agents that predispose to torsades de pointes include Long QT syndromes (often congenital and inherited), hypomagnesemia, hypokalemia, antiarrhythmic drug toxicity (procainamide, flecainide, sotalol, amiodarone), and other drug toxicities (tricyclic antidepressants, calcium channel blockers, phenothiazines). Causes of VT in children include electrolyte disturbances (e.g., hyperkalemia, hypocalcemia, hypomagnesemia), drug toxicity, and acute systemic illnesses or coronary artery abnormalities causing myocardial ischemia.

- II. **DEFINITIONS:**

- A. VT – Ventricular Tachycardia
- B. AHA – American Heart Association

- III. **DEPARTMENT GUIDELINE:**

- A. Assess and manage airway, breathing, and circulation. Monitor ETCO₂ if intubated. Optimize oxygen content.
- B. Establish IV/IO access
- C. Obtain a 12-lead ECG if available. If wide complex (greater than 0.09 seconds), treat as possible VT.
- D. If signs of poor perfusion, administer 0.9% NS or Lactated Ringers 20 ml/kg IV, may be repeated per GUID3203-PR015 – *Intravenous Fluid Therapy*.
- E. If the dysrhythmia is polymorphic VT (Torsade de Pointes), administer **Magnesium sulfate 50 mg/kg IV/IO** as an initial therapy, maximum dose 2 gm, then proceed with traditional VT therapies.
- F. **STABLE PATIENT** (Without signs or symptoms related to the VT):
 - i. Apply appropriately sized pads in preparation for cardioversion or defibrillation.
 - ii. If the patient becomes unstable, proceed to cardioversion as in the Unstable Patient part of this Guideline section.
 - iii. Administer an anti-dysrhythmic.

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- i. **Amiodarone 5 mg/kg IV/IO** over 20-60 minutes (max single dose 150mg). -OR-
- ii. **Procainamide 15 mg/kg IV/IO** over 30-60 minutes. -OR-
- iii. **Lidocaine 1mg/kg IV/IO** repeated every 5 minutes as needed at 0.5 mg/kg IV, until the dysrhythmia is suppressed or 3 mg/kg has been administered. If the dysrhythmia is suppressed, follow with infusion 20 – 50 mcg/kg/min IV.
- iv. Seek expert consultation when giving amiodarone or procainamide. Do not routinely administer amiodarone and procainamide together or with other medications that prolong the QT interval. If these initial efforts do not terminate the rapid rhythm, reevaluate the rhythm.

G. UNSTABLE PATIENT (With signs or symptoms related to the VT):

- i. Consider pre-medication treat pain and/or anxiety per *GUID3203-G005 - Analgesia and Sedation Management*.
- ii. Synchronized cardiovert per *GUID3203-3203 - Cardioversion*.
 - i. Synchronized cardiovert at 0.5 J/kg. Reassess rhythm.
 - ii. Synchronized cardiovert at 1 J/kg. Reassess Rhythm.
 - iii. Synchronized cardiovert at 2 J/kg. Reassess Rhythm, repeat as necessary.
- iii. If the dysrhythmia persists after 4 cardioversions, attempt control with an anti-dysrhythmic as described in the Stable Patient part of this Guideline.
- iv. After conversion, institute anti-arrhythmic therapy as described in the Stable Patient part of this Guideline.
- v. If no response to therapy, contact Medical Control.

H. Always consider the latest recommendations from the American Heart Association.

IV. DOCUMENTATION:

- A. EMS Charts

V. REFERENCES:

- A. Pediatric Advanced Life Support Manual, 2020. American Heart Association.